

KSA Office IT Support & Infrastructure Management

Phase 03: Network Infrastructure Design & Planning

Afaq Business Solutions Co.

Objective

Design a secure, reliable, and scalable office network for Afaq Business Solutions Co. that supports daily business operations, internet access, file sharing, printing, and communication between departments. This phase focuses on planning before implementation in Cisco Packet Tracer.

1. Network Requirements Analysis

- Business networking requirements
- Number of connected devices
- Internet requirements
- Internal communication requirements
- Network security requirements

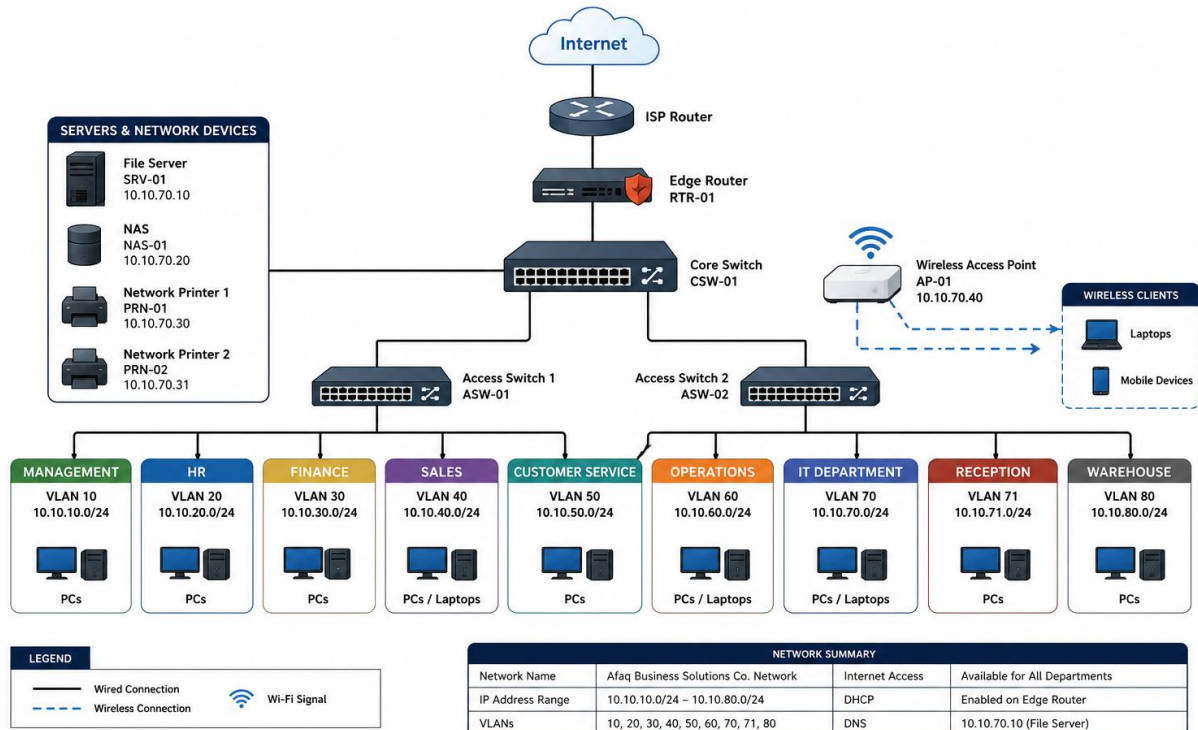
2. Network Topology Design

- ISP
- Edge Router
- Core Switch
- Access Switch 1
- Access Switch 2
- Wireless Access Point
- File Server
- Network Attached Storage (NAS)
- Network Printers
- Desktop Computers
- Laptops

3. Logical Network Diagram

The logical network diagram illustrates the overall network architecture of Afaq Business Solutions Co., showing how all network devices and departments are interconnected. It presents the flow of data from the Internet through the ISP router, edge router, and core switch to the access switches, servers, wireless access point, and end-user devices. The diagram also identifies departmental network segmentation, shared infrastructure, and communication paths to support secure, reliable, and efficient business operations. This design serves as the blueprint for implementing the complete office network in Cisco Packet Tracer.

AFAQ BUSINESS SOLUTIONS CO. LOGICAL NETWORK DIAGRAM



4. IP Addressing Plan

- Network Address
- Subnet Mask
- Default Gateway
- DHCP Range
- Static IP Addresses
- Reserved IP Addresses

5. Department Network Allocation

- Management
- Human Resources
- Finance
- Sales
- Customer Service
- Operations
- IT
- Reception
- Warehouse

6. VLAN Design

VLAN	Department
10	Management
20	HR
30	Finance
40	Sales
50	Customer Service
60	Operations
70	IT
80	Warehouse

7. Network Device Naming Convention

- RTR-01
- CSW-01
- ASW-01
- ASW-02
- AP-01
- SRV-01
- NAS-01
- PRN-01
- PRN-02

8. Network Security Plan

- Device passwords
- VLAN isolation
- Administrative access
- Secure Wi-Fi configuration
- Basic network hardening

9. Network Testing Plan

- End-to-end connectivity
- Internet connectivity
- Inter-VLAN communication
- DHCP address assignment
- DNS resolution
- Printer connectivity
- File server accessibility

10. Cisco Packet Tracer Implementation Plan

- Devices to add
- Cabling requirements
- Interface assignments
- Configuration sequence
- Testing checklist